Induced action graph G as 23 capability families over 154 catalog actions

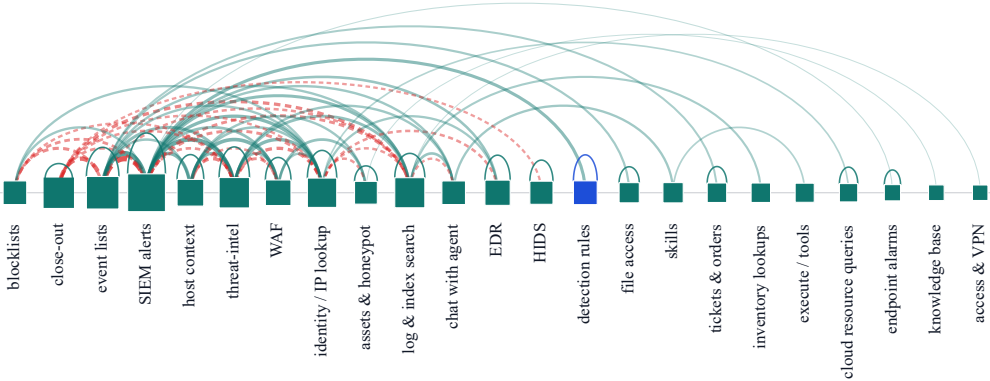
axis order minimises weighted arc span (51,717 to 43,864); square size = episodes, ringed = close-out

solid arcs run with the axis order, dashed against it; colour = dominant domain slice

58 arcs (the 50 heaviest plus 8 so no family is left isolated) carry 64.2% of cross-family mass

the hunting loop is two-directional: SIEM alerts ↔ event lists run both ways

3,709 of the 14,089 advance transitions (26.3%) stay inside their own family; 15,704 advance hops in 1,615 episodes



family	act.	episodes	out / in / intra
SIEM alerts	17	3,825	2,178 / 1,155 / 1,271
event lists	16	2,252	1,556 / 981 / 416
threat-intel	3	1,732	1,050 / 1,139 / 499
log & index search	9	1,660	921 / 656 / 198
identity / IP lookup	11	1,502	890 / 976 / 340
close-out	2	1,126	35 / 1,107 / 6
host context	2	992	707 / 703 / 216
WAF	3	810	615 / 663 / 136
EDR	23	761	368 / 335 / 221
detection rules	13	556	70 / 197 / 265
blocklists	1	537	475 / 533 / 0
chat with agent	1	533	383 / 460 / 0

family	act.	episodes	out / in / intra
HIDS	5	474	268 / 311 / 96
assets & honeypot	8	436	361 / 395 / 6
skills	1	197	120 / 180 / 0
file access	4	190	116 / 158 / 9
tickets & orders	9	141	79 / 122 / 14
inventory lookups	7	139	69 / 134 / 0
execute / tools	4	117	69 / 105 / 0
cloud resource queries	8	76	45 / 60 / 14
endpoint alarms	5	9	4 / 7 / 2
knowledge base	1	2	1 / 2 / 0
access & VPN	1	1	0 / 1 / 0